LAB EXPERIMENT RECORD

Set 28: Vehicle Performance Analysis

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Thursday

Q1. Violin Plot for Fuel Efficiency Across Safety Ratings

Aim: To compare the distribution and variability of fuel efficiency across different safety ratings.

R Code:

R

```
library(ggplot2)

data <- data.frame(

  Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),

  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),

  Horsepower = c(110, 150, 250, 200, 130),

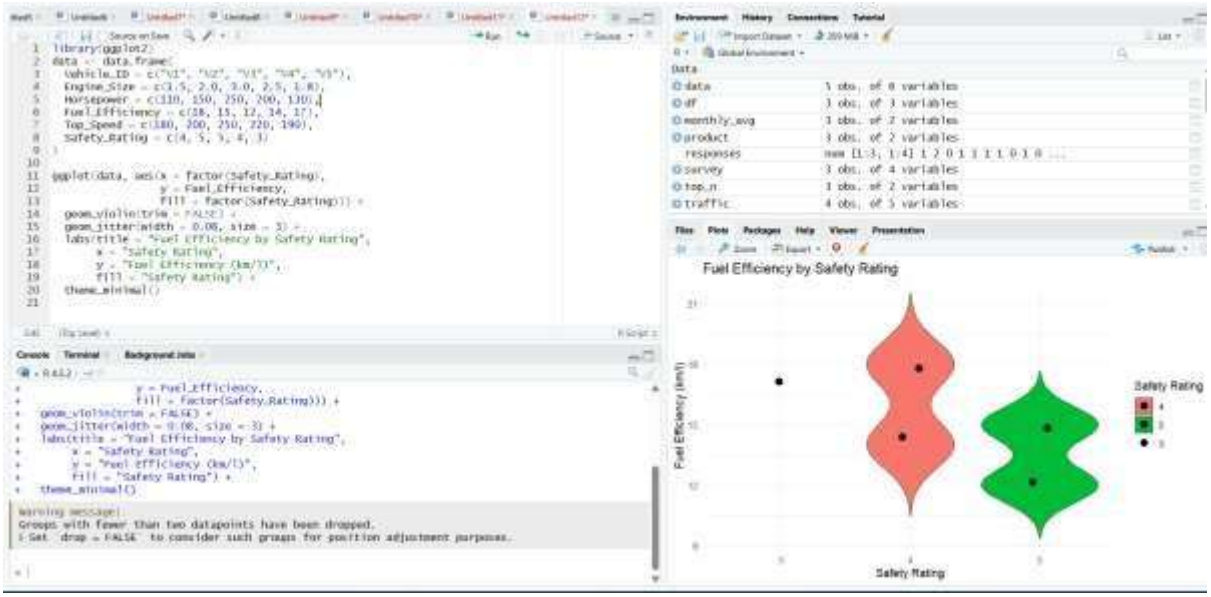
  Fuel_Efficiency = c(18, 15, 12, 14, 17),

  Top_Speed = c(180, 200, 250, 220, 190),

  Safety_Rating = c(4, 5, 5, 4, 3)

)
```

```
ggplot(data, aes(x = factor(Safety_Rating),
                 y = Fuel_Efficiency,
                 fill = factor(Safety_Rating))) +
  geom_violin(trim = FALSE) +
  geom_jitter(width = 0.08, size = 3) +
  labs(title = "Fuel Efficiency by Safety Rating",
       x = "Safety Rating",
       y = "Fuel Efficiency (km/l)",
       fill = "Safety Rating") +
  theme_minimal()
```



Observation: Vehicles with safety rating 3 have fuel efficiency of 17 km/l, while rating 4 vehicles have values of 18 and 14 km/l. Rating 5 vehicles have values of 15 and 12 km/l. The small dataset limits conclusions about the overall distribution and variability.

Result: The violin plot compares fuel efficiency across safety rating categories.

Q2. Scatter Plot of Horsepower and Top Speed

Aim: To examine the relationship between horsepower and top speed using engine size as the color variable.

R Code:

R

```
library(ggplot2)

data <- data.frame(

  Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),

  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),

  Horsepower = c(110, 150, 250, 200, 130),

  Fuel_Efficiency = c(18, 15, 12, 14, 17),

  Top_Speed = c(180, 200, 250, 220, 190),

  Safety_Rating = c(4, 5, 5, 4, 3)

)

ggplot(data, aes(x = Horsepower, y = Top_Speed,

                 color = Engine_Size)) +

  geom_point(size = 4) +

  geom_smooth(method = "lm", se = FALSE,

              color = "black") +

  labs(title = "Horsepower vs Top Speed",

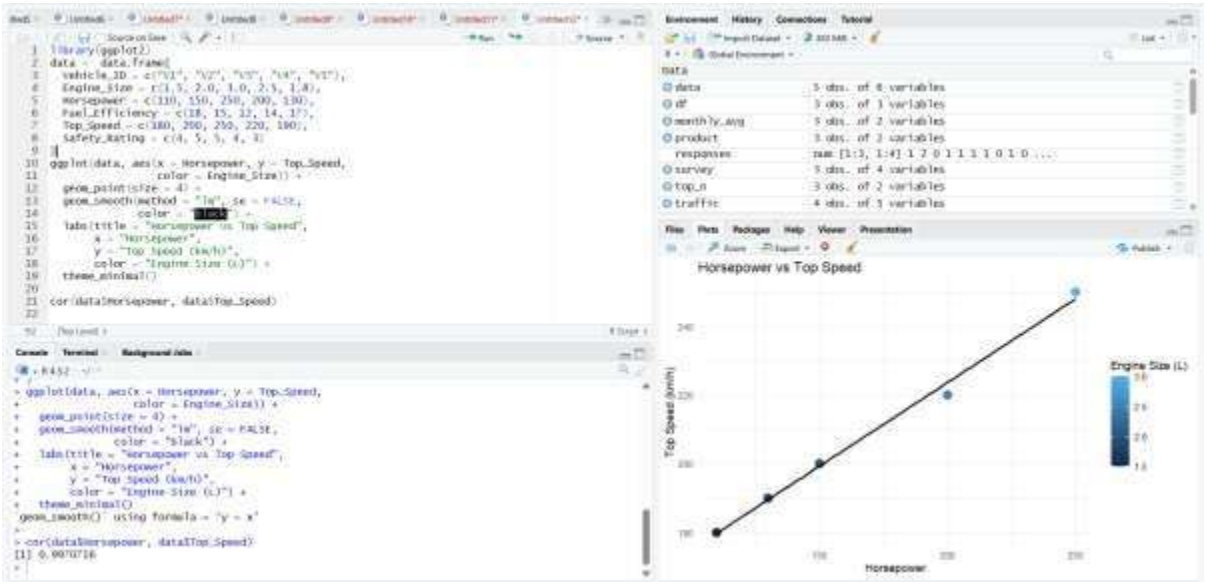
       x = "Horsepower",

       y = "Top Speed (km/h)",

       color = "Engine Size (L)") +

  theme_minimal()
```

cor(data$Horsepower, data$Top_Speed)



Observation: Horsepower and top speed show a strong positive correlation, approximately **0.99**. Vehicles with higher horsepower generally have higher top speeds. V3 has the highest horsepower (250) and top speed (250 km/h).

Result: The scatter plot reveals a strong positive relationship between horsepower and top speed in this dataset.

Q3. Correlation Heatmap for Numerical Variables

Aim: To visualize correlations among numerical vehicle performance variables.

R Code:

R

library(ggplot2)

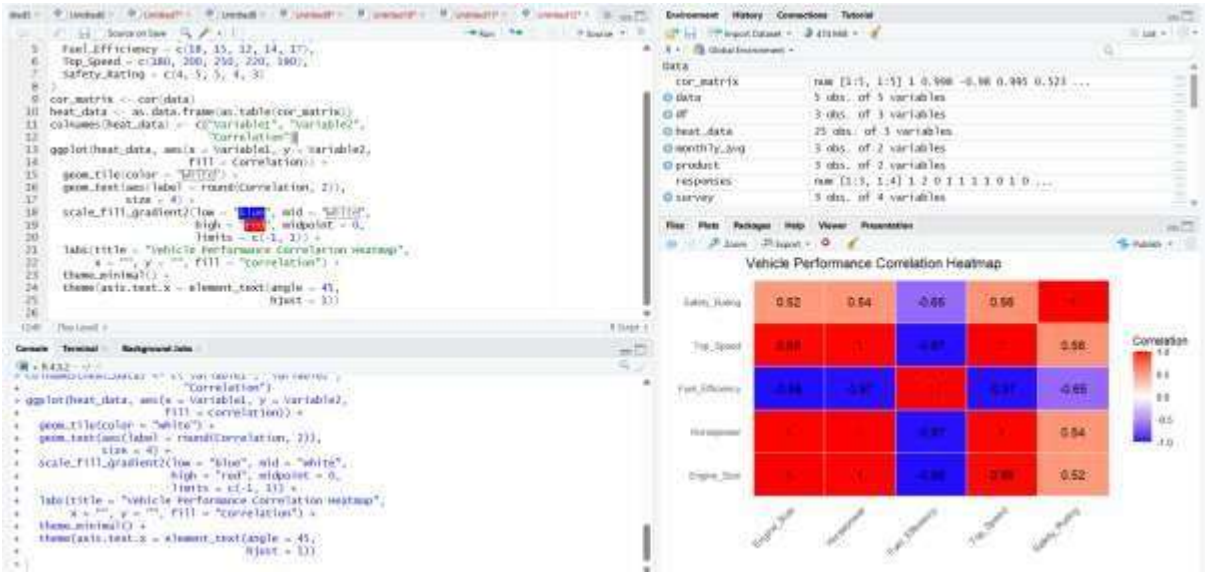
data <- data.frame(

  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),

  Horsepower = c(110, 150, 250, 200, 130),

  Fuel_Efficiency = c(18, 15, 12, 14, 17),

```
  Top_Speed = c(180, 200, 250, 220, 190),
  Safety_Rating = c(4, 5, 5, 4, 3)
)
cor_matrix <- cor(data)
heat_data <- as.data.frame(as.table(cor_matrix))
colnames(heat_data) <- c("Variable1", "Variable2",
                         "Correlation")
ggplot(heat_data, aes(x = Variable1, y = Variable2,
                      fill = Correlation)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(Correlation, 2)),
            size = 4) +
  scale_fill_gradient2(low = "blue", mid = "white",
                       high = "red", midpoint = 0,
                       limits = c(-1, 1)) +
  labs(title = "Vehicle Performance Correlation Heatmap",
       x = "", y = "", fill = "Correlation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45,
                                   hjust = 1))
```



Observation: Horsepower and engine size are positively associated with top speed. Fuel efficiency tends to decrease as engine size, horsepower, and top speed increase. These patterns describe the five supplied vehicles and do not establish causation.

Result: The heatmap displays the strength and direction of relationships among the numerical variables.

Q4. Tableau Interactive Dashboard

Aim: To create a dashboard using bar charts, scatter plots, and filters to identify vehicles balancing fuel efficiency and performance.

Procedure:

1. Enter the given dataset into Excel and save it as Vehicle_Performance.xlsx.
2. Open Tableau and connect to the Excel file.
3. Create a **bar chart** with Vehicle_ID and Fuel_Efficiency.
4. Create a **scatter plot** with Horsepower on Columns and Top_Speed on Rows. Use Engine_Size for color and Vehicle_ID for labels.
5. Add filters for Engine_Size and Safety_Rating.
6. Create a dashboard and combine both charts. Show the filters and tooltips containing vehicle ID, engine size, horsepower, fuel efficiency, top speed, and safety rating.

Vehicle Performance Analysis Dashboard

Engine Size Range (L):

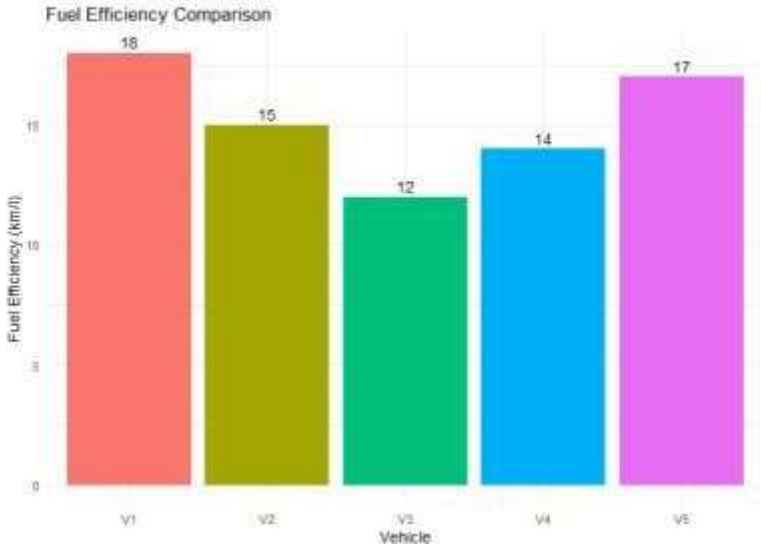
1.5 3

1.5 1.7 1.9 2.1 2.3 2.5 2.7 2.9 3

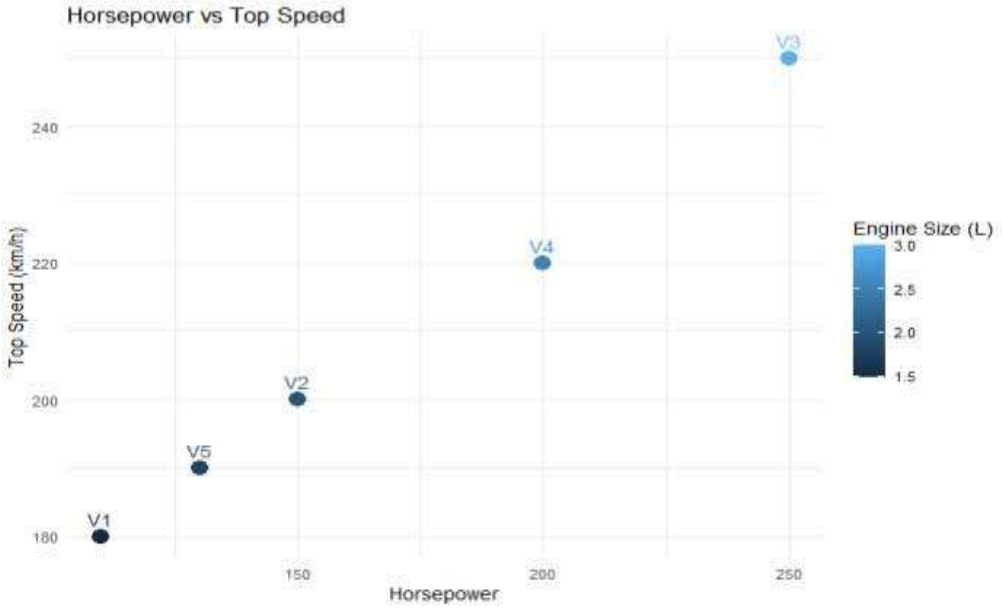
Safety Rating:

All

Fuel Efficiency by Vehicle



Horsepower vs Top Speed



Filtered Vehicle Details

Vehicle_ID	Engine_Size	Horsepower	Fuel_Efficiency	Top_Speed	Safety_Rating
V1	1.50	110.00	18.00	180.00	4.00
V2	2.00	150.00	15.00	200.00	5.00
V3	3.00	250.00	12.00	250.00	5.00
V4	2.50	200.00	14.00	220.00	4.00
V5	1.80	130.00	17.00	190.00	3.00

Analytical Interpretation:

- V1 provides the highest fuel efficiency at 18 km/l, with a top speed of 180 km/h.
- V3 achieves the highest top speed of 250 km/h and horsepower of 250, but its fuel efficiency is the lowest at 12 km/l.
- V2 provides a balance with 15 km/l fuel efficiency, a top speed of 200 km/h, and a safety rating of 5.
- V4 delivers 220 km/h top speed and 14 km/l fuel efficiency, with a safety rating of 4.
- Filters help users compare vehicles based on engine size and safety rating to identify options that suit their performance and efficiency requirements.

Result: The Tableau dashboard supports interactive comparison of vehicle efficiency, speed, horsepower, and safety ratings. The best balance depends on the user's priorities rather than one universally best vehicle.